Van Phan

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Education

Georgia State University *Atlanta, GA*

B.S. Mathematics, GPA: 4.11/4.30 Expected Graduation: May. 2027

- *Relevant Coursework*: Linear Algebra, Differential Equations, Numerical Analysis, Analysis, Modern Algebra, Optimization, Calculus, Biostatistics, Probability and Statistics, Applied Combinatorics, Computer Science

Research Projects

Density and BI-RADS - Aware Triage and Automatic Report Generation for Mammography

- Developed the **Density and BI-RADS-Aware Triage and Report Generation (DB-ATRG)** framework to automate diagnostic text generation and enable risk-based case prioritization, flagging scans with **extremely dense breasts (ACR Category D)** for supplemental screening.
- Fine-tuned the **4-billion parameter MedGemma 1.5** vision-language model using **QLoRA** on the **DMID and VinDr-Mammo dataset**, achieving an **ACR Density Accuracy of 0.7039**, a **ROUGE-L score of 0.8650**, and a **METEOR score of 0.9001** for clinical report generation.
- Accelerated clinical review in simulations on a 100-case cohort by applying a **Cumulative Urgency Score**, capturing **100% of high-risk malignancies** (BI-RADS 4 and 5) within the first 20% of the reading workload (compared to 40% in FIFO) and shifting the **mean rank of severe cases from 42.8 down to 3**.

Modeling Wound Healing Without Migration and Proliferation

- Established a mathematical framework for Age-related Macular Degeneration (AMD) therapeutics, addressing a leading cause of vision loss by modeling Retinal Pigment Epithelium (RPE) wound healing.
- Simulated purse-string contraction and cell fusion repair mechanisms in Python-based CompuCell3D, by utilizing a **Q-Potts cellular model** and **Monte Carlo evaluation** of the **Hamiltonian free energy states**.
- Quantified wound closure efficiency across algorithmic iterations by analyzing texture tensors and spatial points, proving that lower cellular volume constraints, higher wound's radius and affected area yield statistically slower wound closure.

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging

- Developed an automated end-to-end machine learning pipeline for classifying Retinal Pigment Epithelial (RPE) cell during aging, generating insights for potential therapeutic strategies in Age-related Macular Degeneration.
- Achieved highly accurate classification as measured by **90%+ cross-validation F1-score**, by extracting **133 morphological and texture-based features** from 326 RPE cell images and implementing a **stacking ensemble** of **XGBoost, LightGBM, and CatBoost** with a Logistic Regression meta-learner.
- Identified critical RPE aging biomarkers, as demonstrated by top feature importance rankings, suggesting **Local Binary Patterns (LBP)** and **geometric shape metrics** (circularity, aspect ratio) as the most influential classification variables.

Research Paper

- **[In Review] DB-ATRG: The Density and BI-RADS-Aware Triage and Automatic Report Generation System for Mammography**, doi: 10.64898/2026.07.22.26358655.

Posters & Presentations

Density and BI-RADS - Aware Triage and Automatic Report Generation System for Mammography,

- Poster presented at **SIAM Conference on Mathematics of Data Science**, 2026.
- Poster presented at Georgia State Undergraduate Research Conference, 2026.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion,

- **Best Project Award** at Summer Undergraduate Research Symposium, 2025.

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging,

- Poster presented at Georgia State Undergraduate Research Conference, 2025.

Modeling Wound Healing Without Migration and Proliferation,

- Poster presented at Summer Undergraduate Research Symposium, 2024.

Research and Work Experience

RIMMES Program Sep. 2025 – Present

Undergraduate Researcher (Advisor: Dr. Russell Jeter) *Atlanta, GA*

- Led a 3-member student team in DB-ATRG project, developing a system to automatically report clinical report and prioritize high-risk cases in mammography.

Georgia State University Aug. 2025 – Present

Chemistry Laboratory Teaching Assistant *Atlanta, GA*

- Mentored 50+ undergraduate students in Survey of Chemistry I and II, improving understanding of fundamental principles and reducing procedural errors through one-on-one guidance and demonstration of proper laboratory techniques.

Center for the Advancement of Students and Alumni Jun. 2024 – August. 2025

Undergraduate Researcher (Advisor: Dr. Yi Jiang) *Atlanta, GA*

- Led a 3-member student team in two concurrent computational biology projects, developing wound healing simulations and morphological analysis of Retinal Pigment Epithelial (RPE) cells.

Near-peer Mentor

- Provided near-peer mentorship and research support to students in the Math Path Program, guiding research technique implementation and project development for the Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion project.

Selected Projects

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion

- Engineered a **Physics-Informed Neural Networks (PINNs)** model to simulate collective cancer invasion, resulting in a data-efficient, mesh-free solution for complex **Partial Differential Equations (PDEs)**.
- Developed a framework utilizing **Tensorized Fourier Neural Operators (TFNO)** and **PyTorch's autograd engine** to encode physical laws (PDEs, boundary, and initial conditions) directly into the model's loss function, reducing reliance on labeled data.
- Explored advanced techniques, including a **sequence-to-sequence (Seq2seq) PINO** approach and **Augmented Lagrangian methods (ALM)**, to overcome model failure modes like vanishing gradients and improve solution accuracy and stability.

Laboratory Rearing Model of *Lissachatina fulica* for Acharan Sulfate Extraction & Evaluation

- Established a **laboratory rearing model** of *Lissachatina fulica*, using **garlic essential oil** and ***Lactobacillus casei*** for inhibition of pathogenic bacteria (***E. coli***, ***S. aureus***, and ***Salmonella spp.***).
- Extracted **acharan sulfate** of snail flesh and mucus from *Lissachatina fulica*.
- Investigated anticoagulant and thrombolytic properties of extracted **acharan sulfate**, achieving **89.40% anticoagulant activity** within 30 minutes via **clotting time assay** and **32.09% thrombolytic activity** within 1 hour via **clot lysis assay**.

Honors & Awards

- **Best Project Award** at Summer Undergraduate Research Symposium (2025)
- **President's List** at Georgia State University (2023, 2024, 2025, 2026)
- **Prospect Award (National Level) & First Place (City Level)** at Vietnam Science and Engineering Fair (ViSEF) (2022)
- **Consolation Prize** at Engineering and Science Fair of Ho Chi Minh City University of Technology (2022)
- **Bronze Prize** in Physics at the Ho Chi Minh City Excellent Student Contest (2019)

Technical Skills

- **Programming & Tools**: Python, Julia, R, MATLAB, SQL, Git/GitHub, Linux/CLI, LaTeX, CompuCell3D
- **Scientific & Math Modeling**: Physics-Informed Neural Networks, PDE Modeling, Tensorized Fourier Neural Operators, Augmented Lagrangian Method, Q-Potts Cellular Modeling, Monte Carlo Evaluation
- **ML, LLM & AI**: LLMs, Fine-Tuning, Quantization, Transformers, TRL, Hugging Face, Ensemble Methods
- **Libraries**: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, SciPy, Scikit-image, Pillow, Mahotas, Matplotlib, Seaborn